Module Q12: Why Do Models Give Confident Answers Even When Wrong?

Calibration, Hallucination, and the Limits of Softmax Confidence

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The student's intuition — and why it's sharp

The complaint: a model states a wrong fact (or a wrong answer) with no hedging.

- **Confidence** here can mean two different things:
  - *Linguistic* confidence — fluent, assertive phrasing ("The capital is X.").
  - *Probabilistic* confidence — the model's internal probability on its output tokens.
- These are only loosely coupled. A model can emit assertive prose while assigning mediocre internal probability, or vice versa.
- The core issue: **a model's output probability is not a faithful estimate of being correct**. This is the *calibration* problem [1].

Calibration: what "confidence" should mean

A model is **calibrated** if, among all predictions made with confidence p, a fraction p are actually correct.

$$\mathbb{P}(\hat{Y} = Y \mid \hat{P} = p) = p \quad \forall p \in [0, 1].$$

- Measured by *Expected Calibration Error* (ECE): bin predictions by confidence, compare accuracy vs. mean confidence per bin [1].
- Guo et al. [1] showed modern deep networks are systematically **overconfident** relative to older, shallower models — accuracy improved but calibration degraded.
- So "confident when wrong" is a measurable, reproducible property, not a vibe.

Why training produces overconfidence

- **Cross-entropy with hard labels** pushes probability mass toward 1 on the target token. With high-capacity models and no error signal left, the optimizer keeps sharpening logits.
- Softmax has no built-in notion of "I don't know" — probability is always redistributed over the vocabulary, summing to 1.
- **RLHF / instruction tuning** can worsen this: human raters prefer decisive, helpful-sounding answers, so preference optimization rewards assertive phrasing independent of correctness [2].
- Result: linguistic confidence is reinforced even where epistemic warrant is absent.

Hallucination: confident fabrication

- A fluent language model is a *next-token* predictor; it optimizes plausibility of continuations, not truth [3].
- When a fact is absent or underdetermined in context, the most *probable* continuation is often a well-formed but false statement — these are **hallucinations** [4].
- The model has no separate "retrieval failed" pathway; it fills the gap with the statistically typical answer, delivered in the same assertive register as true answers.
- Ji et al. [4] distinguish intrinsic (contradicts source) vs. extrinsic (unverifiable) hallucination — both arrive sounding equally certain.

Do large models even know they're uncertain?

Mixed, encouraging evidence:

- Kadavath et al. [5] find large models can be trained/prompted to predict their own correctness ("P(True)") with non-trivial calibration — the signal exists internally but is not surfaced by default.
- Lin et al. [6] show models can be taught to *verbalize* calibrated uncertainty ("I'm 70% sure").
- Caveat: RLHF-tuned chat models are often *worse* calibrated in raw probabilities than their base counterparts (OpenAI's GPT-4 report notes calibration degrading after RLHF) [7].
- Takeaway: the capacity for self-assessment is partly present but routinely masked by the training objective.

Mitigations

- **Post-hoc calibration:** temperature scaling — divide logits by T tuned on held-out data; cheap and effective for classification [1].
- **Elicited confidence:** prompt the model to state and justify uncertainty; fine-tune to verbalize probabilities [6].
- **Self-consistency:** sample multiple reasoning chains; agreement correlates with correctness [8].
- **Retrieval grounding:** supply authoritative context so the probable continuation *is* the correct one, reducing fabrication.
- **Abstention objectives:** train/evaluate for "I don't know" as a valid, rewarded output.

Answering the student directly

- **Confidence $\neq$ correctness.** Output probability and assertive tone are optimized for plausibility and human preference, not truth.
- **Overconfidence is a training artifact**, exacerbated by hard-label cross-entropy, softmax normalization, and preference tuning.
- **Hallucinations fill knowledge gaps** with typical-sounding text, inheriting the default confident register.
- **The uncertainty signal often exists internally** but is not exposed — it can be recovered through calibration, elicitation, sampling, or grounding.
- Practical rule: treat model fluency as *zero evidence* of accuracy; verify, ground, and request calibrated confidence explicitly.

References

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[4] Z. Ji et al. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), 2023.

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[5] S. Kadavath et al. Language models (mostly) know what they know. *arXiv:2207.05221*, 2022.

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